Final Project Proposal

Project Title

FridgeMate AI – Smart Food Management and Waste Reduction Assistant

Project Overview

FridgeMate AI is a smart web/mobile application designed to help users manage the food stored in their refrigerators, reduce food waste, and make better use of the ingredients they already have at home. The platform combines inventory management, artificial intelligence, and recipe recommendation systems to create a practical and user-friendly solution for everyday food organization.

Many households purchase food that is later forgotten and eventually thrown away due to expiration. FridgeMate AI addresses this problem by allowing users to easily track food items, receive expiration reminders, and generate meal suggestions based on available ingredients.

Motivation for Choosing This Project

I chose this project because food waste is a common real-world problem that affects many households. The idea combines artificial intelligence with a practical application that provides clear value to users.

This project also offers an opportunity to demonstrate multiple AI capabilities, including:

- Object and food recognition
- Natural language processing
- Intelligent recommendation systems
- Automated notifications and decision support

In addition, the project has strong commercial potential and could be expanded into a real startup product in the future.

Project Objectives

The main objectives of FridgeMate AI are:

1. Help users organize and monitor food inventory.

2. Reduce food waste by tracking expiration dates.
3. Provide personalized recipe recommendations.
4. Improve grocery planning and food consumption habits.
5. Demonstrate practical applications of artificial intelligence in everyday life.

Main Features

1. Food Inventory Management

Users can manually enter food items or scan products using their phone camera.

The system stores:

- Food name
- Quantity
- Purchase date
- Expiration date

2. AI Food Recognition

Users can upload an image of groceries or refrigerator contents.

Using AI image recognition, the system identifies food items and automatically adds them to the inventory database.

3. Expiration Tracking and Alerts

The system continuously monitors expiration dates and notifies users when products are close to expiring.

Example:

- "Milk expires in 2 days."
- "Yogurt expires tomorrow."

4. AI Recipe Recommendation Engine

The application analyzes available ingredients and generates meal suggestions.

Example:

Available ingredients:

- Eggs

- Tomatoes
- Cheese

Suggested meals:

- Omelet
- Shakshuka
- Cheese and tomato toast

5. Smart Shopping List Generator

When inventory becomes low or items expire, the system automatically generates a shopping list for the user.

6. Food Waste Analytics Dashboard

The platform provides statistics such as:

- Number of items consumed
- Number of items wasted
- Estimated money saved
- Food waste reduction percentage

Artificial Intelligence Components

The project will utilize AI technologies in several areas:

Computer Vision

Used for recognizing food products from uploaded images.

Possible technologies:

- OpenAI Vision API
- Google Vision API
- TensorFlow image classification models

Natural Language Processing (NLP)

Used for:

- Recipe generation
- Ingredient analysis
- User interaction

Possible technologies:

- OpenAI GPT models

Recommendation System

Used to match available ingredients with suitable recipes and meal suggestions.

Proposed Technology Stack

Frontend

- React.js
- HTML
- CSS
- JavaScript

Backend

- Node.js
- Express.js

Database

- MongoDB

AI Services

- OpenAI API
- Computer Vision API

Deployment

- Vercel or Netlify (Frontend)
- Render or Railway (Backend)

System Workflow

1. User adds food items manually or through image scanning.
2. Items are stored in the database.
3. Expiration dates are monitored automatically.
4. AI analyzes available ingredients.
5. The system generates recipes and recommendations.
6. Users receive alerts for expiring products.
7. Analytics dashboard tracks food usage and waste reduction.

Expected Outcomes

At the completion of the project, users will be able to:

- Maintain a digital inventory of refrigerator contents.
- Receive intelligent expiration reminders.
- Generate recipes using available ingredients.
- Reduce food waste.
- Improve grocery management habits.

Conclusion

FridgeMate AI combines artificial intelligence with a practical everyday problem. The project demonstrates the integration of computer vision, recommendation systems, and natural language processing in a single platform. It provides measurable benefits to users while showcasing modern AI technologies and software development skills.